

# THE ZERO PARADOX

*ZP-F: The Counterexamples*

*Version 1.0 | May 2026*

This document is self-contained within ordered field theory. No p-adic topology, no information theory, and no Hilbert space machinery is required. All results follow from the axioms of a linearly ordered field alone.

Illustrated Companion: A paired ZP-F Illustrated Companion provides plain-language explanation and motivation for the results here. Examples and intuitions are kept separate from the formal layers to distinguish illustrative material from proofs.

## I. Purpose and Scope

The Zero Paradox framework requires a metric space in which a minimal first departure from the null state is structurally forced — the Binary Snap. This document establishes the negative result: no linearly ordered field can serve as that substrate. The real numbers  $\mathbb{R}$  are the canonical and most familiar instance; the theorem applies to any field carrying a compatible linear order.

The blocking mechanism is elementary: in any such field, for any positive element  $\varepsilon$ , the element  $\varepsilon/2$  exists, is positive, and is strictly smaller. There is no floor. The Binary Snap requires a floor. Therefore the snap is structurally impossible in any linearly ordered field.

This result contextualises the choice of  $Q_2$  in ZP-B:  $Q_2$  is not chosen arbitrarily or for novelty. It is the minimal metric structure in which the structural isolation of zero is built in rather than excluded.

## II. The General Result

The following results hold for any type  $F$  carrying a field structure, a linear order, and the `IsStrictOrderedRing` property — the standard Mathlib replacement for the deprecated `LinearOrderedField` typeclass (deprecated October 2025). All proofs are machine-verified in Lean 4.

| Theorem F-DENSITY                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\forall \varepsilon : F, 0 < \varepsilon \rightarrow 0 < \varepsilon/2 \wedge \varepsilon/2 < \varepsilon$                                                                            |
| For any positive element $\varepsilon$ in a linearly ordered field $F$ , the element $\varepsilon/2$ is also positive and strictly smaller than $\varepsilon$ .                        |
| Proof: <code>half_pos</code> gives $0 < \varepsilon/2$ ; <code>half_lt_self</code> gives $\varepsilon/2 < \varepsilon$ . Both follow from the ordered field axioms. Status: DERIVED. ✓ |
| Theorem F-NO-MIN                                                                                                                                                                       |
| $\neg \exists \varepsilon : F, 0 < \varepsilon \wedge \forall \delta : F, 0 < \delta \rightarrow \varepsilon \leq \delta$                                                              |
| No linearly ordered field has a minimal positive element.                                                                                                                              |

#### Theorem F-NO-MIN

Proof: Suppose such  $\varepsilon$  exists. By F-DENSITY,  $0 < \varepsilon/2 < \varepsilon$ . Then  $\varepsilon \leq \varepsilon/2$  (by minimality) contradicts  $\varepsilon/2 < \varepsilon$ . Status: DERIVED. ✓

#### Theorem F-SNAP-BLOCKED

$\forall \varepsilon_0 : F, 0 < \varepsilon_0 \rightarrow \exists \delta : F, 0 < \delta \wedge \delta < \varepsilon_0$

Any candidate first step  $\varepsilon_0 > 0$  from 0 in a linearly ordered field is blocked:  $\varepsilon_0/2$  is a smaller positive element, so  $\varepsilon_0$  cannot be a genuine first step.

Proof: Take  $\delta = \varepsilon_0/2$ . F-DENSITY gives  $0 < \delta < \varepsilon_0$ . Status: DERIVED. ✓

#### Theorem F-SNAP-IMPOSSIBLE

$\neg \exists \varepsilon_0 : F, 0 < \varepsilon_0 \wedge \neg \exists \delta : F, 0 < \delta \wedge \delta < \varepsilon_0$

The Binary Snap cannot occur in any linearly ordered field.

A snap requires a minimal first departure from 0 with nothing below it. F-SNAP-BLOCKED shows such a floor never exists in a linearly ordered field.

Proof: Direct from F-SNAP-BLOCKED. Status: DERIVED. ✓

### III. The Real Numbers — Canonical Instance

The following are corollaries of the general results above, instantiated at  $F = \mathbb{R}$ . They are stated explicitly because  $\mathbb{R}$  is the most familiar and intuitive case, and because the ZP-F companion document is written around  $\mathbb{R}$  as its primary example.

#### Corollary R-DENSITY

$\forall \varepsilon : \mathbb{R}, 0 < \varepsilon \rightarrow 0 < \varepsilon/2 \wedge \varepsilon/2 < \varepsilon$

Density at zero for  $\mathbb{R}$ . Follows from F-DENSITY at  $F = \mathbb{R}$ . Status: DERIVED (corollary). ✓

#### Corollary R-NO-MIN

$\neg \exists \varepsilon : \mathbb{R}, 0 < \varepsilon \wedge \forall \delta : \mathbb{R}, 0 < \delta \rightarrow \varepsilon \leq \delta$

$\mathbb{R}$  has no minimal positive element. Follows from F-NO-MIN at  $F = \mathbb{R}$ . Status: DERIVED (corollary). ✓

#### Corollary R-SNAP-BLOCKED

$\forall \varepsilon_0 : \mathbb{R}, 0 < \varepsilon_0 \rightarrow \exists \delta : \mathbb{R}, 0 < \delta \wedge \delta < \varepsilon_0$

Any candidate first step in  $\mathbb{R}$  can be halved. Follows from F-SNAP-BLOCKED at  $F = \mathbb{R}$ . Status: DERIVED (corollary). ✓

| Corollary R-SNAP-IMPOSSIBLE                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\neg \exists \varepsilon_0 : \mathbb{R}, 0 < \varepsilon_0 \wedge \neg \exists \delta : \mathbb{R}, 0 < \delta \wedge \delta < \varepsilon_0$ |
| The Binary Snap cannot occur in $\mathbb{R}$ . Follows from F-SNAP-IMPOSSIBLE at $F = \mathbb{R}$ . Status: DERIVED (corollary). ✓             |

### IV. Axiom Profile

All eight results (F-DENSITY, F-NO-MIN, F-SNAP-BLOCKED, F-SNAP-IMPOSSIBLE and their  $\mathbb{R}$  corollaries) verify under #print axioms with the profile: propext, Classical.choice, Quot.sound. This is the standard profile for Mathlib field and order results. No snap-specific axioms are introduced.

### V. Relationship to the Framework

ZP-F is self-contained and does not feed formally into any other layer. It is the negative complement to ZP-B: where ZP-B establishes that  $\mathbb{Q}_2$  can host the snap (via the 2-adic valuation and clopen ball structure), ZP-F establishes that  $\mathbb{R}$  — and any linearly ordered field — cannot.

The dependency order of the framework places ZP-F as independent:

| Layer | Role                                                                  | Dependency                        |
|-------|-----------------------------------------------------------------------|-----------------------------------|
| ZP-B  | p-Adic topology — $\mathbb{Q}_2$ hosts the snap                       | ZP-A                              |
| ZP-F  | Counterexample — $\mathbb{R}$ and ordered fields cannot host the snap | None (self-contained)             |
| ZP-G  | Category theory                                                       | Self-contained; ZP-E conceptually |

### VI. Validation Status

| Component         | Status / Notes                                               |
|-------------------|--------------------------------------------------------------|
| F-DENSITY         | DERIVED — Lean-verified; general case                        |
| F-NO-MIN          | DERIVED — Lean-verified; general case                        |
| F-SNAP-BLOCKED    | DERIVED — Lean-verified; general case                        |
| F-SNAP-IMPOSSIBLE | DERIVED — Lean-verified; general case                        |
| R-DENSITY         | DERIVED — Corollary of F-DENSITY at $F = \mathbb{R}$         |
| R-NO-MIN          | DERIVED — Corollary of F-NO-MIN at $F = \mathbb{R}$          |
| R-SNAP-BLOCKED    | DERIVED — Corollary of F-SNAP-BLOCKED at $F = \mathbb{R}$    |
| R-SNAP-IMPOSSIBLE | DERIVED — Corollary of F-SNAP-IMPOSSIBLE at $F = \mathbb{R}$ |